

Project: Cinema Management System.

Group member 04:

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1) Business Process Description:

Business Context (Where): The cinema operates within the entertainment industry, providing movie screenings and additional services (e.g., snacks, beverages) in physical cinema locations. The business aims to offer a smooth ticket booking experience for its guests through an application that helps staff and managers efficiently manage ticket orders, schedules, and customer feedback.

Users:

1. **Manager:** Oversee cinema operations, manage movies, schedules, discounts, and manage user accounts (create/update/delete), troubleshoot system issues, and ensure data integrity. .
2. **Staff:** Handle ticket bookings for guests, assist with movie, seating selection, apply discounts, and add snacks and beverages to orders.

Business Process:

1. **Account Management Process:** Managers create and update user accounts (staff and managers). This ensures access control and up-to-date user information. Each account must have a different username.
2. **Room/seat Management Process:** Managers use the seat editor to create and modify seat structures for each cinema room. Manager also defines the price for each type of seat in that room.
3. **Schedule Management Process:** Whenever a cinema has a new movie, Manager schedules it by the day it starts showing, showtime and show in which room.
4. **Discount Management Process:** Before giving away discounts to the guest, the manager has to define the discount information into the system. Each discount is unique and can be applied only one time and in one ticket.
5. **Snacks/Beverage Management Process:** Manager adds available products into the system with the unique ID to manage and the price for each product.
6. **Ticket Booking Process:** Staff assist guests with selecting a movie, choosing seats, and adding snacks/beverages. Discounts are applied if available, and a booking confirmation is generated. Each ticket may contain more than 1 seat, but it must be in the same schedule, movie, room and cinema. When the booking is completed, mark the seat to be booked to classify and mark Discount to be used. Tickets can not be refunded once the movie starts.
7. **Review/Analysis Process:** After each movie is finished, the system will compute the ratio of seats which are booked and will compute the revenue of each movie within a

day. From that data, the manager can decide whether to keep showing that movie or not, or to keep showing in which frequency.

Features/Characteristics:

1. Movies: Titles, genres, durations, trailers, age ratings.
2. Seats: Seat numbers, layouts for each cinema room, availability status.
3. Snacks/Beverages: Types, prices, and availability.
4. Users: Personal details (name, phone, email, gender), role (manager, staff).
5. Schedules: Schedule movie date, schedule movie start , room assignments..
6. Discounts: Discount ID, date expire,. percentages, eligibility criteria (e.g., specific user groups or pass specific value).

2) Application description

Create/Update Account:

- **Who will use it?:** Manager.
- **Inputs:** Username, name, address, phone number, email, gender.
- **Outputs:** Account creation confirmation or account update details.
- **Description:** Admin creates and updates user accounts (e.g., staff or manager accounts). Fields are validated, such as gender ("M" or "F") and unique usernames.

Book Ticket:

- **Who will use it?:** Staff (on behalf of guests).
- **Inputs:** Movie selection, screening schedule, seat selection, snack/beverage selection, discount application.
- **Outputs:** Ticket confirmation with order details.
- **Description:** Staff assist guests in selecting a movie, seat, and additional services like snacks. Discounts can be applied, and a final booking receipt is generated.

Seat Editor:

- **Who will use it?:** Managers.
- **Inputs:** Room selection, seat structure details (rows, numbers, special sections).
- **Outputs:** Updated seating arrangement.
- **Description:** Managers configure or edit the seat layout for each cinema room. This affects seat availability during the ticket booking process.

Manage Movies:

- **Who will use it?:** Managers.
- **Inputs:** Movie title, genre, duration, trailer link,description, some information about this movie and actor.
- **Outputs:** Updated movie listings.

- **Description:** Managers add new movies, edit or delete existing ones, and assign screening times and rooms for each movie.

Manage Discounts:(optional)

- **Who will use it?:** Managers.
- **Inputs:** Discount name, percentage, validity dates, user criteria.
- **Outputs:** Discount available during ticket booking.
- **Description:** Managers create, update, or delete discounts and set specific conditions for their application (e.g., VIP users or certain showtimes).

Manage Schedule:

- **Who will use it?:** Managers.
- **Inputs:** Movie title, room, schedule time start, date.
- **Outputs:** Updated screening schedules.
- **Description:** Managers assign screening times to each movie and ensure no room overlaps. This affects ticket availability for each movie.

Check revenue of one movie:

- **Who will use it?:** Managers.
- **Inputs:** Date_start, Date_from, Movie_id.
- **Outputs:** The total revenue from this period of time.
- **Description:** Managers can check the revenue of a movie in some days so that they can make an analysis of this and may change the schedule of this movie.

Compute ratio booked ticket:

- **Who will use it?:** Managers.
- **Inputs:** Schedule_id.
- **Outputs:** The ratio of booked seats overall total seats in one schedule.
- **Description:** Managers can see the cover ratio to make a decision to remain showing that movie or show in lower or higher frequency.

3) ERD and DB schema:



